

One sided rectangulations

Logarithmic power and non D finiteness

OEIS A348351

$$a(n) = \Gamma^n n^{-\alpha+o(1)},$$

$$\Gamma = \frac{7 + \sqrt{17}}{2}, \quad \alpha = 1 + \frac{\pi}{\arccos((-29 + 7\sqrt{17})/4)}.$$

This report proves the exponential growth constant and the logarithmic polynomial exponent for one-sided, equivalently area-universal, rectangulations. It also proves that their ordinary generating function is not D-finite and obtains the corresponding two-term threshold inverse.

The stronger multiplicative equivalent $a(n) \sim c\Gamma^n n^{-\alpha}$, the existence or value of a positive amplitude c , and correction or transseries expansions are not established here.

[write] **Status and trust boundary, added 5 October 2026, batch 103.** This is bundle Report 111, filed in ProveIt as a single-source research report (provenance, credits, relation to the repository, notation and the collected non-claims in Section 1.1). It is AI-assisted, unrefereed and not formalized; its title page and PDF metadata name no person (author “Research report”). The walk model is the Asinowski–Cardinal–Felsner–Fusy bijection [2], used and not re-proved; the martingale functional limit theorem [3] and the André–Chudnovsky–Katz theorem in the form of [4] are cited, and ProveIt’s intake did not check the cited page locations. The theorem proves the exponent part of the conjecture of [2]; its amplitude part is open. Four condensed steps are marked where they occur; the write completes two of them, and the other two are open (Section 11.4).

2 October 2026

Self-contained argument with exact finite checks and reproducible source

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1 Results and their scope

Let $a(n)$ be OEIS A348351, with $a(0) = 1$. For $n \geq 1$ it counts one-sided rectangulations with n rectangles, also called area-universal rectangulations. Equivalently, it counts permutations of $[n]$ avoiding the four vincular patterns

$$2\text{-}41\text{-}3, \quad 3\text{-}14\text{-}2, \quad 2\text{-}14\text{-}3, \quad 3\text{-}41\text{-}2.$$

In each pattern the middle two positions must be adjacent; these are not ordinary classical pattern-avoidance conditions [1, 2].

Define

$$\Gamma = \frac{7 + \sqrt{17}}{2}, \quad \rho = \frac{29 - 7\sqrt{17}}{4}, \quad \theta = \arccos(-\rho), \quad p = \frac{\pi}{\theta}, \quad \alpha = 1 + p. \quad (1.1)$$

Numerically $\Gamma \approx 5.561552812808830$ and $\alpha \approx 2.9569294595087534$. All arguments below use the exact expressions.

Theorem 1.1. *As $n \rightarrow \infty$,*

$$\log a(n) = n \log \Gamma - \alpha \log n + o(\log n), \quad a(n) = \Gamma^n n^{-\alpha+o(1)}. \quad (1.2)$$

In particular $\lim_{n \rightarrow \infty} a(n)^{1/n} = \Gamma$. The ordinary generating function $F(z) = \sum_{n \geq 0} a(n)z^n$ is not D-finite over $\mathbb{C}(z)$. If $N(y) = \min\{n \geq 0 : a(n) \geq y\}$, then, as $y \rightarrow \infty$,

$$N(y) = \frac{\log y}{\log \Gamma} + \frac{\alpha}{\log \Gamma} \log \log y + o(\log \log y). \quad (1.3)$$

The published result in Asinowski, Cardinal, Felsner and Fusy [2, Proposition 4.23 and the following paragraph, printed pp. 39–40] gives the upper exponential rate Γ and conjectures

$$a(n) \sim c \Gamma^n n^{-\alpha} \quad (c > 0). \quad (1.4)$$

Theorem 1.1 determines its exponential rate and its logarithmic power, and proves the stated non-D-finiteness consequence without assuming (1.4). It does not settle the amplitude part of that conjecture. For example, a factor $\exp(o(\log n))$ need not converge. There is no claimed asymptotic correction series, all-orders expansion, or transseries in this report.

Here “not D-finite” means that no nonzero linear differential equation with polynomial coefficients annihilates F . Equivalently, $a(n)$ has no fixed-order linear recurrence with polynomial coefficients valid for all sufficiently large n . This does not rule out other kinds of formulas or recurrences: the multistate lattice recurrence in Section 2 remains exact and effective.

Proof plan. The source bijection supplies a finite-color quadrant walk. A Perron transform makes it a centered Markov-additive process. An explicit bounded martingale correction gives a Brownian scaling limit. We prove an unrestricted lattice local limit theorem by finite Fourier matrices and derive an exact-time interior bridge estimate. Brownian wedge harmonic measure, transferred uniformly over large annuli, gives the logarithmic escape exponent. Two first-hit prefixes and an exact-time bridge then give the excursion exponent without overcounting. Finally, coefficient positivity and the arithmetic local structure of G-functions imply non-D-finiteness; monotone bracketing gives the inverse.

The color-dependent increments are not i.i.d. No i.i.d. cone-excursion theorem is applied, and no random regeneration time is substituted for the original length. The arithmetic and martingale limit theorems used are explicitly identified with their hypotheses. The finite checks supplement, but do not prove, the probability limits.

1.1 Provenance, credits and reading conventions

[write] **Added 5 October 2026, batch 103.** *Provenance.* This report is Report 111 of the session bundle that ProveIt’s docs/incoming intake processed as batch 103: the archive `A348351_Rectangulations_Exponent_and_Non_D_Finiteness_Source.zip` (436,299 bytes; wrapper directory `report111/`, 23 files), arrival commit `60f54ea06`, placed in `9c995cefe`. The text is the delivered `report111.tex`, shipped as `article.tex`. The delivered 18-page PDF `report111.pdf`, the integrity manifests `manifest.json` and `checks/CHECKSUMS.sha256`, and two JSON files that are byte copies of shipped ones are not shipped; all are retrievable from the arrival commit. The package names no ProveIt commit, path or report, so the report has no pin; its title page and PDF metadata name no person (author “Research report”), and it carries no “prepared for private review” line. It is a single-source report, so no merge choice was made. The write prefixed the 55 delivered labels with `osr`: (“one-sided rectangulations”), added the notes marked [write], this subsection with Remark 1.2, Remarks 9.1 and 10.1, and Section 11.4; it suppressed the page anchors of the title page (which shared the anchor `page.1` with the first arabic-numbered page) and set the bibliography ragged right (one underfull line). No statement, proof or number of the source was changed. Theorem and equation numbers follow the section counter (Theorem 1.1 is Theorem 1.1).

Inherited material, as the manuscript credits it. (i) The walk model of Section 2 is the bijection of Asinowski, Cardinal, Felsner and Fusy [2, Section 4.6, Figure 4.16] between one-sided rectangulations and weak left/right four-colour quadrant walks; it is used, not re-proved. ProveIt’s intake did not consult [2]; it reconstructed the step table from the two weak inequalities of Section 2 and recomputed $a(0), \dots, a(16)$, which agree with the OEIS. (ii) The upper exponential rate Γ is [2, Proposition 4.23], and (1.4) is the conjecture stated after it. Theorem 1.1 proves the *exponent part* of that conjecture, $\log a(n) = n \log \Gamma - \alpha \log n + o(\log n)$; the *amplitude part*, the existence of $c > 0$ with $a(n) \sim c\Gamma^n n^{-\alpha}$, remains open (Question 1). (iii) The martingale functional limit theorem is Whitt’s [3]; the arithmetic local structure of G-functions is the André–Chudnovsky–Katz theorem in the formulation of [4]; [5] is cited for comparison only. The page locations of these citations are the source’s readings.

Relation to the repository. Before batch 103 no file of the repository mentioned A348351, one-sided or area-universal rectangulations, or the Asinowski–Cardinal–Felsner–Fusy paper, so the source’s bounded literature statement (Section 11) needed no correction. The report meets the following repository texts.

- `a377922-corner-polyhedra-schnyder` (same collection), Part I. Its Lemma `cps:lem:nonD` proves non-D-finiteness from two-sided bounds $a_n = \Theta(\mu^n n^{-\alpha})$, α irrational. The criterion of Section 10 needs only $a(n) = \Gamma^n n^{-\alpha+o(1)}$, a strictly weaker hypothesis; Remark 10.1 states it in general, so that `cps:lem:nonD` is its special case. The mechanism (a positive derivative of order $k > \alpha - 1$, then rational local exponents of G-functions) is the same; the extra step is the $\pm \varepsilon$ room in the exponent. The irrationality argument of Section 10 (its first subsection) differs from that of `cps:lem:irrational`: there $2 \cos \theta$ is a rational non-integer ($9/8, 44/27$), hence not an algebraic integer; here $u = 2 \cos \theta = (-29 + 7\sqrt{17})/2$ is an algebraic integer (a root of $u^2 + 29u + 2$), so that route is closed, and the certificate is instead its conjugate $u' < -2$ (Kronecker: every conjugate of $2 \cos(2\pi j/m)$ lies in $[-2, 2]$).
- The same report’s Part III, Report 112 of this bundle: see Remark 1.2.
- The threshold inverse (1.3) is, at its precision, an instance of `p0:thm:lambert-core` of the transseries volume `transseries_and_inversion.tex` in the directory `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/` (Remark 9.1); no novelty is claimed for the inversion mechanics.
- `a113226-vincular-avoiders` treats a different vincular pattern (12-34); no shared question or theorem.

No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status.

Remark 1.2 ([write] The cone theorem of Report 112). Report 112 of the same bundle (the same session and day; the bundle index records 08:51 UTC for Report 111 and 09:26 UTC for Report 112), filed by batch 103 as Part III of `a377922-corner-polyhedra-schnyder` in this collection (file prefix `47-logcone-`, labels `cps:log::`; written concurrently with this report), runs the method of

Sections 3–8 as a general theorem. Its Theorem 2.1 (`cps:log:thm:cone`) concerns a translation-invariant Markov-additive chain on $\mathbb{Z}^2 \times E$, E finite, with (H1) a primitive phase matrix P ; (H2) a uniform exponential moment of the jumps; (H3) zero stationary drift and a positive definite effective covariance Σ ; (H4) Fourier spectral radius below one at every nonzero point of the torus and a simple unit eigenvalue at zero; and (H5) for endpoint phases a, b : for every R , a legal positive-probability path from $(0, a)$, and one for the stationary dual from $(0, b)$, reaching whitened radius at least R with whitened angle in a fixed compact interval $J \subset (0, \theta)$. It concludes $K_n((0, a), (0, b)) = n^{-1-\nu+o(1)}$ with $\nu = \pi/\theta$, $\theta = \arccos(-\rho)$ and ρ the correlation of Σ .

The chain of Section 3 satisfies (H1)–(H5) for $(a, b) = (c, W)$, $c \in \{B, R, G\}$. (H1): P has the zero pattern of A , and $A^2 > 0$; π in (3.3) is positive. (H2): every step has coordinates in $\{-1, 0, 1\}$. (H3): $\sum_c \pi_c m(c) = 0$, and Σ in (3.7) is positive definite; Report 112 normalizes the corrector by $\sum_c \pi_c h(c) = 0$ instead of $h(W) = 0$, which changes h by a constant vector and leaves every corrected increment $v + h(d) - h(c)$, hence Σ , unchanged. (H4) is Section 4: zero is the only point of spectral radius one, and there the unit eigenvalue is simple since P is primitive. (H5): the seed of Section 8, H east then H north steps from $(0, B)$ in colour B , and its dual counterpart from $(0, W)$ (the reversals of the $W \rightarrow W$ steps $-E, -N$, with dual probability $p_{WW}^*(-v) = p_{WW}(v) > 0$), end at (H, H) on the bisector, at angle $\theta/2 \in J = [\theta/3, 2\theta/3]$ and whitened radius $H |\Sigma^{-1/2}(1, 1)| \rightarrow \infty$; from $(0, R)$ or $(0, G)$ first take the zero-displacement edge to B (column B of the table). Hence $K_L((0, c), (0, W)) = L^{-1-p+o(1)}$ for $c \in \{B, R, G\}$, with $\nu = p$ since both are $\pi/\arccos(-\rho)$ for the same Σ . For $c = W$, $K_L((0, W), (0, W)) = 0$ for every $L \geq 1$: every edge leaving W has displacement $-E$ or $-N$ (row W of the table) and leaves Q from the origin. With (3.12) and $r_c > 0$ this gives $a(L+1) = \Gamma^L L^{-1-p+o(1)}$, which is (1.2).

So the exponent of Theorem 1.1 also follows from Report 112's theorem. The present report is the earlier, model-specific proof; its bounded steps make the annular transfer simpler (Report 112 handles exponential-tail jumps with relative-overshoot and bad-jump estimates). Its non-D-finiteness argument is the same as Report 112's Section 5. Neither report cites the other.

Notation (a table of the letters the manuscript reuses; no symbol was renamed).

Symbol	Meanings (where)	Not to be read as
Γ	the growth constant $(7 + \sqrt{17})/2$, Perron root of A	the Gamma function (which does not occur)
α, p	$\alpha = 1 + p$ the coefficient exponent, $p = \pi/\theta$ (1.1); $p_{cd}(v)$ edge probabilities (3.2), p_{dc}^* (3.10)	p as a probability; in a377922 Part I, $p = \alpha - 1$ too, but Report 112 writes ν
ρ, θ	correlation of Σ , $\theta = \arccos(-\rho)$ the whitened wedge angle	
s	$s = \sqrt{17}$ (Section 3); survival $s_n(c)$, $s_n^*(c)$ (7.7); the real exponent s in (10.1)	s_n as a power of $\sqrt{17}$
c, C	a colour $c \in \mathcal{C}$; positive constants $c, C, c_B, C_B, C_\eta, c_\eta, C_\varepsilon$; the amplitude c of (1.4) and the leading coefficient c of (10.3); C_j the colour process; $\mathcal{C}$ the colour set	the conjectured amplitude as any proved constant
B, R, G, W	the four colours (sans-serif)	B_Σ (Brownian motion), R, G_n , the Lambert branch W_{-1} (Remark 9.1)
E, N	the steps $(1, 0)$, $(0, 1)$; $N(y)$ the threshold (1.3) and $N = N(y)$ in Section 9; $\mathcal{E}_H$ a path event (Section 7)	N in Section 9 as a step
R	a radius scale (Lemma 3.1, Section 7); $R = \Gamma^{-1}$ the radius of convergence (Section 10); R the colour; R_0, R_1 radii	

Symbol	Meanings (where)	Not to be read as
J	the central arc $[\theta/3, 2\theta/3]$ (Sections 6–8); the number of annular crossings, with r_J (Sections 7, 8)	the crossing count as the arc
T	the time cutoff $T(q)$ (Sections 6–8); the Fourier matrix $T(t)$ (Section 4)	
H	the seed height (Section 8); $H(t) = F(Rt)$ (Section 10); the harmonic function H_f (6.1); the horizon H of $\mathcal{E}_H$	
L	the walk length $L = n - 1$ (Sections 2, 3, 8); $L = \log y$ (Section 9)	
q	the annulus ratio (Sections 6–8); a rational exponent $q \in \mathbb{Q}$ (10.3)	
A	the matrix $A = M(1, 1)$; a compact set A_0 (Lemma 5.1); constants A, A_B	
b, B	the one-step whitened bound b ; coefficients b_k (6.1); constants b_0, b_B ; the interval end b of Lemma 5.1; a colour b (Section 8); B_Σ Brownian motion	
a	the sequence $a(n)$; the interval end a of Lemma 5.1; a colour a (Section 8); in Remark 1.2, Report 112’s endpoint phases a, b	
F, G	the generating function F ; prefix masses F_n, G_n (8.3); $\mathbf{G}$ the colour	F_n as a coefficient of F
u	$u = 2 \cos \theta$, u' its conjugate; the eigenvector in (4.1); $t = e^{-u}$ in the proof of (10.1); a dual prefix length	
β, η, δ	β the disk-confinement constant and $\beta = 3 - \alpha$ (Section 10); η the exponent tolerance; δ the scale of the last radius	

[write] **What is not claimed, collected from the source.** No multiplicative equivalent $a(n) \sim c\Gamma^n n^{-\alpha}$, no positive amplitude, no correction series, no transseries; “a factor $\exp(o(\log n))$ need not converge”. Non-D-finiteness excludes only a fixed-order polynomial-coefficient differential equation (equivalently recurrence) and does not rule out the exact multistate recurrence (2.2); no claim is made about arbitrary non-arithmetic D-finite functions. The inverse’s constant $-(\alpha/g) \log g$ “lies below the proved error scale” and is not claimed. Lemma 5.1 is not an endpoint limit theorem at the cone vertex; no survival equivalent or limiting constant is asserted; no killed-cone local limit theorem, no i.i.d. cone theorem and no random regeneration time is used. The finite checks do not prove the functional or local limit theorems, the harmonic measure, the annular transfer, bridge tightness, the logarithmic theorem, G-function regularity or the non-D-finiteness deduction; agreement with the 17 OEIS terms is not a proof of the source bijection; the annular example parameters of the check suite are illustrative. [5] is cited for comparison only. The literature search was bounded, and the result “should be compared with subsequent specialist work before any publication or priority claim”. The delivered manifest is an integrity record, not a digital signature. No third-party PDF is redistributed. The write adds: the source’s readings of the literature are not checked by ProveIt, and its numerical observations (Section 11.4) are uncertified floating point.

2 The exact colored walk model

Set $E = (1, 0)$, $N = (0, 1)$ and $\mathcal{C} = \{\mathbf{B}, \mathbf{R}, \mathbf{G}, \mathbf{W}\}$, in this order. Write $Q = \mathbb{R}_{\geq 0}^2$. For $n \geq 1$, the source bijection identifies $a(n)$ with length $L = n - 1$ walks in $\mathbb{Z}^2 \cap Q$, starting at $(0, 0)$ in any color, and ending at $(0, 0)$ in color $\mathbf{W}$. Every visited point must be in Q . The allowed edges are as follows; two entries in a cell mean two distinct displacement choices.

Start \ end	B	R	G	W
B	E, N	E, N	E, N	E, N
R	0	0	$0, E - N$	$0, E - N$
G	0	$0, N - E$	0	$0, N - E$
W	$\emptyset$	$-E$	$-N$	$-E, -N$

This is the weak leftright history-walk model of [2, Section 4.6, Figure 4.16]. The weak leftmost and rightmost conditions, rather than their strong counterparts, must be intersected. Their logical “or” conditions produce this table. The initial color is unrestricted; imposing an initial white color would change the model.

For an algebraic encoding define S_{cd} to be the cell of displacements for colors c, d . One can reconstruct the table independently by putting

$$\ell(\mathbf{B}) = 1, \quad \ell(\mathbf{R}) = \ell(\mathbf{G}) = 0, \quad \ell(\mathbf{W}) = -1,$$

requiring $v_x + v_y = \ell(c)$ and the two lower bounds

$$v_x \geq \begin{cases} 0, & c \in \{\mathbf{B}, \mathbf{R}\} \text{ or } d \in \{\mathbf{B}, \mathbf{G}\}, \\ -1, & \text{otherwise,} \end{cases}$$

$$v_y \geq \begin{cases} 0, & c \in \{\mathbf{B}, \mathbf{G}\} \text{ or } d \in \{\mathbf{B}, \mathbf{R}\}, \\ -1, & \text{otherwise.} \end{cases}$$

The vertical quantity $v_x + v_y$ is prescribed by the departure color, not by the arrival color. These conditions describe this specific source model, not arbitrary rectangulation classes.

The unrestricted step matrix is

$$M(x, y) = \begin{pmatrix} x+y & x+y & x+y & x+y \\ 1 & 1 & 1+x/y & 1+x/y \\ 1 & 1+y/x & 1 & 1+y/x \\ 0 & x^{-1} & y^{-1} & x^{-1}+y^{-1} \end{pmatrix}, \quad A = M(1, 1). \quad (2.1)$$

Here A^2 is strictly positive, so A is primitive, and

$$\det(tI - A) = t(t+1)(t^2 - 7t + 8).$$

Thus Γ is its Perron root.

An exact counting recurrence. Let $u_0(x, y, c) = \mathbf{1}_{(x,y)=(0,0)}$ for each $c \in \mathcal{C}$. Extend all u_j by zero outside Q , and put

$$u_{j+1}(x, y, d) = \sum_{c \in \mathcal{C}} \sum_{v \in S_{cd}} u_j(x - v_x, y - v_y, c), \quad (x, y) \in \mathbb{Z}_{\geq 0}^2. \quad (2.2)$$

Then $a(L+1) = u_L(0, 0, \mathbf{W})$. The check suite reproduces the source prefix

$$a(0), \dots, a(16) =$$

$$1, 1, 2, 6, 20, 72, 274, 1088, 4470, 18884, 81652, 360054,$$

$$1614618, 7346688, 33856008, 157777908, 742637416.$$

It also compares against direct permutation enumeration, separately encoding the middle adjacency rule. These finite agreements are useful model checks, not a proof of the source bijection.

3 The Perron chain and effective covariance

Put $s = \sqrt{17}$. A positive right eigenvector for Γ is

$$r = \left(\frac{1+s}{2}, \frac{3+s}{4}, \frac{3+s}{4}, 1 \right)^T. \quad (3.1)$$

For each individual allowed edge $c \rightarrow d$ with displacement v , give that edge probability

$$p_{cd}(v) = \frac{r_d}{\Gamma r_c}. \quad (3.2)$$

Set it to zero on forbidden edges. Since $Ar = \Gamma r$, the outgoing probabilities sum to one. The color transition matrix is

$$P = \Gamma^{-1} \text{diag}(r)^{-1} A \text{diag}(r),$$

a primitive stochastic matrix. Its stationary row vector is

$$\pi_B = \pi_W = \frac{51-5s}{136}, \quad \pi_R = \pi_G = \frac{17+5s}{136}. \quad (3.3)$$

All entries are positive and sum to one; direct multiplication gives $\pi P = \pi$.

Let C_j be the color and S_j the spatial position of the unrestricted chain. The conditional mean displacement $m(c)$ is

$$\begin{aligned} m(B) &= (1/2, 1/2), & m(W) &= (-1/2, -1/2), \\ m(R) &= ((s-3)/4, (3-s)/4), & m(G) &= ((3-s)/4, (s-3)/4). \end{aligned}$$

In particular $\sum_c \pi_c m(c) = 0$. The following explicit vector-valued function solves the Poisson equation $(I - P)h = m$:

$$\begin{aligned} h(B) &= ((7+s)/8, (7+s)/8), \\ h(R) &= ((-21+13s)/32, (5+3s)/32), \\ h(G) &= ((5+3s)/32, (-21+13s)/32), \\ h(W) &= (0, 0). \end{aligned} \quad (3.4)$$

Every coordinate of h lies in $[0, 3/2)$. Consequently

$$Z_j = S_j + h(C_j) - h(C_0) \quad (3.5)$$

is a martingale, because the conditional expectation of its increment is $m(c) + (Ph)(c) - h(c) = 0$. Its increments have coordinate absolute values below 3.

The long-run covariance is the stationary average of the corrected increment outer products:

$$\Sigma = \sum_{c,d} \sum_{v \in S_{cd}} \pi_c p_{cd}(v) [v + h(d) - h(c)][v + h(d) - h(c)]^T. \quad (3.6)$$

Substitution gives the exact matrix

$$\Sigma = \frac{1}{272} \begin{pmatrix} 51+13s & -17+5s \\ -17+5s & 51+13s \end{pmatrix}. \quad (3.7)$$

Its eigenvalues are $(34 + 18s)/272$ and $(68 + 8s)/272$, both positive. Its correlation is precisely ρ in (1.1), and $0 < \rho < 1/2$.

One independent algebraic check uses the spectral polynomial

$$\begin{aligned} \det(tI - M(x, y)) &= t[t^3 - U(x, y)t^2 + t + V(x, y)], \\ U(x, y) &= x + y + 2 + x^{-1} + y^{-1}, \\ V(x, y) &= x + y + x/y + y/x + 2 + x^{-1} + y^{-1}. \end{aligned} \tag{3.8}$$

If $\lambda(x, y)$ is the analytic eigenvalue with $\lambda(1, 1) = \Gamma$, then differentiating this identity shows that the Hessian at zero of $\log \lambda(e^u, e^v) - \log \Gamma$ is (3.7); its gradient is zero. This is the cumulant covariance of the additive process and agrees with (3.6).

Lemma 3.1 (Unrestricted functional limit). *For every initial color, with polygonal interpolation,*

$$R^{-1}(S_{\lfloor R^2 t \rfloor} - S_0) \implies B_\Sigma(t) \tag{3.9}$$

on every compact time interval as $R \rightarrow \infty$, where B_Σ is Brownian motion with covariance matrix Σ per unit time. The convergence is uniform over the finite set of initial colors.

Proof. For Z in (3.5), the predictable quadratic variation is the sum of the conditional corrected increment covariance matrices. The ergodic theorem for the finite primitive color chain makes this sum, divided by R^2 , converge to $t\Sigma$. Bounded increments imply that scaled jumps tend uniformly to zero, that the expected maximum squared scaled jump tends to zero, and that jumps of the scaled predictable quadratic variation tend to zero. Thus the martingale functional central limit theorem, in the predictable-covariation form of [3, Theorem 2.1(ii), pp. 270–271], applies. The bounded difference $Z_j - S_j$ disappears under scaling. Polygonal and step interpolation differ by $O(R^{-1})$. Finitely many initial colors give the stated uniformity. Translation invariance also permits varying starting positions whose scaled limits exist. $\square$

[write] **Condensed step 1 of 4, completed here; added 5 October 2026.** The proof states Whitt's conditions in prose. The detail it leaves implicit is that the predictable quadratic variation is an additive functional of the colour chain. Given the past up to time j , with $C_j = c$, the increment of Z is $w = v + h(d) - h(c)$ with probability $p_{cd}(v)$; its conditional mean is $m(c) + (Ph)(c) - h(c) = 0$, and its conditional covariance is $V(c) = \sum_d \sum_{v \in S_{cd}} p_{cd}(v) ww^T$, a function of c alone. So the predictable quadratic variation of $R^{-1}Z$ at time t is $R^{-2} \sum_{j < \lfloor R^2 t \rfloor} V(C_j)$. The ergodic theorem for the finite irreducible chain C makes it converge almost surely, from every initial colour, to $t \sum_c \pi_c V(c)$, which is $t\Sigma$ by (3.6). The jumps of $R^{-1}Z$ are at most $3\sqrt{2}/R$ in norm, and those of the predictable quadratic variation at most $\max_c \|V(c)\|/R^2$, deterministically. These are the hypotheses of the predictable-covariation martingale functional limit theorem. What stays external is that theorem itself, as the source reads [3]; ProveIt did not check the citation.

3.1 Reversal and the exact count identity

Define the dual unrestricted chain by

$$p_{dc}^*(-v) = \frac{\pi_c}{\pi_d} p_{cd}(v). \tag{3.10}$$

Stationarity makes its outgoing probabilities sum to one. Its color matrix

$$P^* = \text{diag}(\pi)^{-1} P^T \text{diag}(\pi)$$

is primitive. Its stationary drift is zero. A bounded Poisson corrector exists on the stationary-mean-zero subspace. Reversing a stationary length- j path negates its total displacement, so its long-run covariance is again Σ . The argument of Lemma 3.1 therefore applies to the dual chain with the same limit.

Let $K_j((x, c), (y, d))$ be the probability of going between the displayed states in j steps while all visited locations remain in Q , and define K_j^* for the dual. Edgewise telescoping gives

$$\pi_c K_j((x, c), (y, d)) = \pi_d K_j^*((y, d), (x, c)). \quad (3.11)$$

Every specified length- L original path from color c to W has probability $1/(\Gamma^L r_c)$. Consequently

$$a(L+1) = \Gamma^L \sum_{c \in \mathcal{C}} r_c K_L((0, c), (0, W)). \quad (3.12)$$

In particular $a(n) \leq (3+s)\Gamma^{n-1}$ for $n \geq 1$. It remains to prove the polynomial logarithmic exponent of these killed endpoint probabilities.

4 An unrestricted lattice local limit theorem

Define the Fourier matrix

$$T(t)_{cd} = \sum_v p_{cd}(v) e^{it \cdot v}, \quad t \in [-\pi, \pi]^2.$$

Its spectral radius is at most one. Suppose it has an eigenvalue z with $|z| = 1$. The equality case in the triangle inequality, propagated through the irreducible color matrix, forces an eigenvector u to have equal nonzero coordinate moduli and gives

$$e^{it \cdot v} u_d = z u_c \quad (4.1)$$

on every allowed edge. The zero $R \rightarrow R$ loop gives $z = 1$.

There are red-return paths of common length three with displacements $0, E, -E, N, -N$:

- Three red zero loops have displacement 0.
- $R \rightarrow B$ with displacement 0, then $B \rightarrow R$ with E or N , then a red zero loop give E or N .
- $R \rightarrow W$ with 0, then $W \rightarrow R$ with $-E$, then a red zero loop give $-E$.
- $R \rightarrow W$ with 0, then $W \rightarrow G$ with $-N$, then $G \rightarrow R$ with 0 give $-N$.

Multiplying (4.1) along these paths forces $e^{it_1} = e^{it_2} = 1$. Thus zero is the only point of spectral radius one on the Fourier torus. This verifies both space and time aperiodicity; it is stronger than merely observing that the coordinate increments generate $\mathbb{Z}^2$.

At zero, the simple eigenvalue is 1 with projection $\mathbf{1}\pi$. Finite-dimensional analytic perturbation, zero drift, and (3.7) give

$$\lambda_T(t) = 1 - \frac{1}{2} t^T \Sigma t + O(|t|^3), \quad \Pi_T(t) \rightarrow \mathbf{1}\pi.$$

The other eigenvalues are uniformly separated from one near zero. On the complement of a small neighborhood of zero, the preceding strict spectral-radius bound and compactness give uniform exponential decay of matrix powers. This last assertion also holds for nonnormal matrices: choose

a common spectral contour strictly inside the unit disk and use the resolvent formula, or a finite covering of power-norm bounds. Near zero, $|\lambda_T(t)| \leq e^{-c|t|^2}$.

[write] Condensed step 2 of 4, completed here; added 5 October 2026. The uniform decay off a neighbourhood U of zero, written out by the second route. Let $\|\cdot\|$ be the maximum-row-sum norm. Since $|T(t)_{cd}| \leq \sum_v p_{cd}(v) = P_{cd}$ and P is stochastic, $\|T(t)^j\| \leq 1$ for all t and $j \geq 0$. On the compact set $K = [-\pi, \pi]^2 \setminus U$ every $T(t)$ has spectral radius below one, so by Gelfand's formula each $t \in K$ has an integer $k_t \geq 1$ with $\|T(t)^{k_t}\| < 1$; by continuity of $s \mapsto T(s)^{k_t}$ there are $\delta_t > 0$ and an open neighbourhood V_t of t with $\|T(s)^{k_t}\| \leq 1 - \delta_t$ for $s \in V_t$. Finitely many V_t cover K ; let k and δ be the largest k_t and the smallest δ_t among them. For $s \in V_t$ and $j \geq 0$, submultiplicativity and the bound 1 on the remaining power give $\|T(s)^j\| \leq (1 - \delta)^{\lfloor j/k_t \rfloor} \leq (1 - \delta)^{j/k-1}$. So $\sup_{s \in K} \|T(s)^j\|$ decays exponentially in j , with no normality assumption.

Fourier inversion, split into this neighborhood and its complement, now proves the local limit theorem. In the neighborhood put $t = u/\sqrt{j}$ and use dominated convergence. The error is bounded by its Fourier L^1 norm independently of the oscillatory spatial factor, hence uniformly in displacement:

$$\sup_{z \in \mathbb{Z}^2, c, d} \left| j \mathbb{P}_c(S_j - S_0 = z, C_j = d) - \pi_d g_\Sigma(z/\sqrt{j}) \right| \rightarrow 0, \quad (4.2)$$

where

$$g_\Sigma(w) = \frac{1}{2\pi\sqrt{\det \Sigma}} \exp(-w^T \Sigma^{-1} w/2).$$

For completeness, the analytic eigenvalue expansion is the characteristic-function version of the cumulant expansion following (3.8); bounded corrections do not alter its second derivative. Enlarging a constant to cover finitely many small times yields

$$\sup_{x, y, c, d} \mathbb{P}_{(x, c)}((S_j, C_j) = (y, d)) \leq C/j \quad (j \geq 1). \quad (4.3)$$

The dual Fourier matrix is $\text{diag}(\pi)^{-1} T(-t)^T \text{diag}(\pi)$, so the same statements hold for the dual. No killed-cone local limit theorem has been used.

5 Interior bridges at exact integer times

Lemma 5.1. *Let A_0 be a compact subset of the open quadrant, and let $0 < a < b < \infty$. There exist $c > 0$ and n_0 such that for $n \geq n_0$, every integer $m \in [an, bn]$, every $x, y \in \mathbb{Z}^2$ with $x/\sqrt{n}, y/\sqrt{n} \in A_0$, and every color pair c_0, d_0 ,*

$$K_m((x, c_0), (y, d_0)) \geq c/n. \quad (5.1)$$

Proof. The unrestricted endpoint probability is at least c_1/n by (4.2), uniformly over the specified compact ranges. We show that the unrestricted bridge conditioned on this exact endpoint has a uniformly positive probability of staying in Q .

First prove its Brownian bridge limit. Through time $k = \lfloor tm \rfloor$, for fixed $t < 1$, its density relative to the unconditioned process is

$$\frac{\mathbb{P}_{C_k}(S_{m-k} - S_0 = y - S_k, C_{m-k} = d_0)}{\mathbb{P}_{c_0}(S_m - S_0 = y - x, C_m = d_0)}. \quad (5.2)$$

The numerator is at most $C/(m-k)$ by (4.3) and the denominator at least c/m . Thus the density is uniformly bounded for fixed $t < 1$. On compact scaled spatial sets, (4.2) makes it converge

uniformly to the Gaussian transition-density ratio defining a covariance- Σ Brownian bridge. The terminal color factor π_{d_0} cancels. The unrestricted functional limit, tightness, and this bounded density prove convergence on each $[0, t]$. Contributions outside a compact spatial set are uniformly negligible by unrestricted tightness and the density bound.

The reversed conditioned bridge is exactly the dual conditioned bridge: the path-reversal factor depends only on endpoints and cancels upon conditioning. Apply the same argument to the dual bridge. The forward restriction to $[0, 2/3]$ and the backward restriction to $[1/3, 1]$ are tight. If two times are less than $1/3$ apart, both lie in at least one of these overlapping intervals; their two modulus-of-continuity bounds therefore give tightness on $[0, 1]$. The finite-dimensional limits and fixed terminal point identify the full Brownian bridge limit.

For endpoints in a compact subset of the open quadrant, their joining segments stay uniformly away from the axes. The deviation of a Brownian bridge from its joining segment is a centered Brownian bridge independent of its endpoints. It has positive probability of remaining in a fixed sufficiently thin tube around zero. This gives a uniform positive probability for the desired tube around each joining segment, also uniformly in the allowed duration ratios. The tube condition is open in continuous path space. Portmanteau gives a positive lower bound for the corresponding discrete conditioned probabilities.

All uniform claims can alternatively be proved by contradiction: a failing sequence has convergent scaled endpoints and duration ratios and constant endpoint colors along a subsequence. The preceding limit applies to that subsequence. Convexity of Q means that the polygonal path stays in Q precisely when its lattice vertices do. Multiplying by the unrestricted endpoint lower bound proves (5.1). $\square$

This lemma concerns endpoints at distances of order $\sqrt{n}$ from the sides. It is not an endpoint limit theorem at the cone vertex.

[write] Condensed step 3 of 4, a sketch; added 5 October 2026. The proof of Lemma 5.1 is correct in outline (intake reading), and its ingredients are standard, but three pieces are only described: (i) the transfer lemma that turns a uniformly bounded density (5.2), converging uniformly on compacts, together with the unrestricted functional limit and tightness, into weak convergence of the conditioned bridge on $[0, t]$; (ii) the dual functional and local limit theorems and the dual kernel bound, asserted in Section 3 and at the end of Section 4 by the symmetry of the construction rather than written out; (iii) the uniformity over $m \in [an, bn]$, endpoints and colours, which rests on the one-paragraph subsequence argument. Writing these out is Question 6.

6 Brownian wedge crossings

Whiten spatial positions by $X_j = \Sigma^{-1/2} S_j$. The Brownian limit becomes standard planar Brownian motion. The image of Q is a closed wedge K whose angle is

$$\theta = \arccos(-\rho).$$

Indeed the cosine of the angle between $\Sigma^{-1/2} E$ and $\Sigma^{-1/2} N$ is $-\rho$. After a rotation,

$$K^\circ = \{(r, \phi) : r > 0, 0 < \phi < \theta\}.$$

Its central angular interval is $J = [\theta/3, 2\theta/3]$. Coordinate interchange symmetry of Σ makes the ray through $\Sigma^{-1/2}(1, 1)$ the bisector.

For Brownian motion in the wedge truncated at radius $q > 1$, the probability of exiting through an outer-arc set with indicator f is the bounded harmonic function

$$H_f(r, \phi) = \sum_{k \geq 1} b_k (r/q)^{kp} \sin(kp\phi), \quad b_k = \frac{2}{\theta} \int_0^\theta f(\psi) \sin(kp\psi) d\psi, \quad p = \pi/\theta. \quad (6.1)$$

This follows by separation of variables in Laplace's equation: the sine modes vanish on the wedge sides and the radial powers give the boundary Fourier sine series on the circle. Bounded-domain uniqueness identifies the harmonic measure. At $r = 1$ the series is absolutely convergent. The finitely many indicator discontinuities and corners have zero Brownian exit probability and do not change the formula.

For $f = 1$, the coefficients are $4/(k\pi)$ for odd k and zero for even k . For $f = \mathbf{1}_J$, $b_1 = 2/\pi > 0$ and $|b_k| \leq 2$. For $\phi \in J$, $\sin(p\phi) \geq \sqrt{3}/2$. Thus the positive first mode dominates the $O(q^{-2p})$ tail for all sufficiently large q . There exist $q_B > 1$ and positive constants c_B, C_B , independent of $q \geq q_B$, such that

$$\sup_{\phi \in [0, \theta]} \mathbb{P}_{(1, \phi)}(\text{reach radius } q \text{ before leaving } K) \leq C_B q^{-p}, \quad (6.2)$$

$$\inf_{\phi \in J} \mathbb{P}_{(1, \phi)}(\text{exit at radius } q \text{ inside } J) \geq c_B q^{-p}. \quad (6.3)$$

At a wedge-side start the first probability is zero, because the normal one-dimensional Brownian projection visits the exterior immediately. The harmonic expression extends continuously to zero on either side.

There are also universal disk exit-time constants. From any starting point in a disk of radius q , an increment of norm greater than $3q$ at time q^2 forces exit by that time. Brownian scaling gives a positive lower bound independent of the point and q . Iteration gives constants $A_B, b_B > 0$ such that

$$\mathbb{P}(\text{stay in the radius-}q \text{ disk up to } Tq^2) \leq A_B e^{-b_B T}. \quad (6.4)$$

For each fixed q , choose finite $T(q)$ large enough that the probability in (6.3), with the extra condition of hitting before $T(q)q^2$, is still at least $(c_B/2)q^{-p}$. A time cutoff can be chosen at a continuity point, or enlarged slightly.

7 Uniform annular transfer and survival

The next arguments apply to both the original and the dual process, with common constants. Let b bound the norm of a single whitened displacement in either process.

7.1 Disk confinement

There are $\beta > 0$ and $R_1 < \infty$ such that for $R \geq R_1$, every color and every point in the radius- R disk,

$$\mathbb{P}(\text{stay in that disk for } \lceil R^2 \rceil \text{ steps}) \leq 1 - \beta.$$

To see this, the event $|X_{\lceil R^2 \rceil} - X_0| > 3R$ forces an exit and has probability uniformly bounded below by Lemma 3.1. Its law is independent of X_0 by translation invariance. The Markov property yields

$$\mathbb{P}_x \left(\max_{j \leq t} |X_j| \leq R \right) \leq (1 - \beta)^{\lfloor t/\lceil R^2 \rceil \rfloor} \leq A \exp(-b_0 t/R^2). \quad (7.1)$$

The last bound holds after changing absolute constants for $R \geq R_1$.

7.2 Transfer including limiting starts on the sides

Fix $q \geq q_B$. Start at a lattice state with whitened radius in $[R, R + b]$. Let σ be its first discrete time at radius at least qR and let τ be its first discrete time outside the closed wedge. There are constants $c, C > 0$, independent of sufficiently large q , followed by choices $T = T(q) < \infty$ and $R_0 = R_0(q) < \infty$, for which every $R \geq R_0$ satisfies

$$\mathbb{P}_x(\sigma < \tau) \leq Cq^{-p} \quad \text{for every initial angle in } K, \quad (7.2)$$

$$\mathbb{P}_x(\sigma < \tau, \arg X_\sigma \in J, \sigma \leq T(qR)^2) \geq cq^{-p} \quad \text{for every initial angle in } J. \quad (7.3)$$

A successful event requires $\sigma < \infty$. The radial overshoot is at most b .

Here are the needed uniformity details. Rescale space by R and time by R^2 . Any sequence of admissible starting positions has a subsequence tending to a point of radius one, and initial colors can be fixed along a further subsequence.

For a fixed finite horizon H , the continuous-path event

$$\mathcal{E}_H = \{w : \exists t \leq H, |w(t)| \geq q, w([0, t]) \subseteq K\}$$

is closed in the uniform topology. Given convergent paths and witness times, pass to a convergent subsequence of those times; continuity and closedness of K preserve both conditions. A successful discrete trajectory before the horizon has its polygonal interpolation in this event because K is convex. For Brownian paths, (6.2) bounds its probability. If the limiting initial point lies on a side, its probability is zero: a normal Brownian projection crosses outside arbitrarily soon, while the outer radius cannot be reached at time zero. Touching a side before an outer hit adds no contribution, by the same immediate-crossing property after the first touch.

Portmanteau and the compact-subsequence argument therefore give the upper bound uniformly over starts, including those only a bounded number of lattice steps from a side, for hits by time Tq^2 after scaling. Later hits have probability at most the disk-confinement tail (7.1), bounded by Ae^{-b_0T} . Choose $T(q)$ so this is at most q^{-p} , then $R_0(q)$ so the FCLT error costs at most one further q^{-p} . This proves (7.2), for example with $C = C_B + 2$. Floors and ceilings can be absorbed by a slightly enlarged fixed time horizon.

For the lower bound, use Brownian paths whose first exit is through the interior of the outer arc J before Tq^2 . Increase $T(q)$ so their probability is at least $(c_B/2)q^{-p}$. This is an almost-sure continuity event for Brownian motion from the compact interior starting arc. The exit time is finite, the path hits none of the finitely many corners or arc endpoints, and it crosses the smooth exit circle immediately. On a successful path it stays a positive distance from both wedge sides up to and just beyond its exit, by continuity on this compact time interval. Exceptional boundary points are polar for planar Brownian motion. An exit time has no atom at a positive deterministic time, since exit exactly then would place a Gaussian position on the boundary.

Equivalently, restrict first to successful paths with terminal angle in a smaller open subarc and time strictly below the cutoff, then take open uniform neighborhoods of such paths. Vanishing one-step radial and angular overshoots guarantee a discrete exit in J for all sufficiently close paths. Portmanteau gives (7.3) with, say, $c = c_B/4$. Compactness of the initial arc and finiteness of the colors give uniformity by the same subsequence argument.

These transfers use the raw positions, not a fictitious fixed boundary for the corrected martingale. They require no equality of color-conditioned covariances, only the common effective covariance and unrestricted limit.

[write] **Condensed step 4 of 4, a sketch; added 5 October 2026.** The uniformity of (7.2) and (7.3) over starting radii in $[R, R + b]$, angles (including starts a bounded number of steps from a side) and colours is argued by contradiction and compactness in prose. Correct in outline (intake reading). Missing is the written argument: a failing sequence of starts, radii and colours; a subsequence with converging scaled start and fixed colour; the functional limit from a converging scaled start (Lemma 3.1 and translation invariance); and portmanteau for the closed event $\mathcal{E}_H$ (upper bound) and for an open set of uniform neighbourhoods of successful Brownian paths (lower bound), where the discrete-to-continuous comparison uses that the scaled one-step overshoot b/R tends to zero. Also Question 6.

7.3 Logarithmic escape and survival bounds

Fix $\eta \in (0, p)$. Select a single sufficiently large $q \geq \max(2, q_B)$ such that

$$Cq^{-p} \leq q^{-p+\eta}, \quad cq^{-p} \geq q^{-p-\eta}. \quad (7.4)$$

Only after choosing q select $T(q)$ and $R_0(q)$ as above, and enlarge R_0 so that

$$(q - 1)R_0 > b. \quad (7.5)$$

Fix $r_0 \geq R_0$ and put $r_j = q^j r_0$. At a first hit of r_j the radius belongs to $[r_j, r_j + b]$; (7.5) ensures the next radius cannot already have been skipped. The strong Markov property and (7.2) show, uniformly over colors at the origin,

$$\mathbb{P}_0(\text{reach radius } R \text{ before leaving } K) \leq C_\eta R^{-p+\eta}, \quad R \geq r_0. \quad (7.6)$$

Indeed the initial reach of r_0 costs at most one, and reaching r_J requires J successful crossings. For an arbitrary R , take J with $r_J \leq R < r_{J+1}$ and absorb the fixed factors into C_η .

Let $s_n(c) = \mathbb{P}_{(0,c)}(\tau > n)$. A surviving path either reaches R or stays in its disk for n steps. Hence

$$s_n(c) \leq C_\eta R^{-p+\eta} + A \exp(-b_0 n / R^2).$$

Take $R = \sqrt{n / (M \log n)}$, where fixed M is sufficiently large to make the second term at most a constant times n^{-p-1} . Logarithmic factors in the first term are $n^{o(1)}$. Since η is arbitrary, for every $\varepsilon > 0$,

$$\max_c s_n(c) \leq C_\varepsilon n^{-p/2+\varepsilon}. \quad (7.7)$$

The same bound holds for dual survival $s_n^*(c)$. No survival equivalent or limiting constant is asserted.

8 Time bounded escapes and exact excursion gluing

8.1 Fixed seeds and accepted first hit prefixes

There are fixed positive-probability seeds from $(0, B)$ in the original chain and from $(0, W)$ in the dual chain to (H, H) with the respective colors unchanged. Take H east steps followed by H north steps: $B \rightarrow B$ allows both original steps, and the reversals of $W \rightarrow W$ west and south steps allow both dual steps.

Choose the integer H large enough that $r_0 = |\Sigma^{-1/2}(H, H)| \geq R_0$. This seed ends on the bisector, inside J . It stays at whitened radius at most r_0 . To verify the last point, its squared whitened norm is a positive multiple of

$$x^2 + y^2 - 2\rho xy.$$

This convex quadratic takes its maximum on $[0, H]^2$ at a vertex, and the value at (H, H) exceeds those at $(H, 0)$ and $(0, H)$ because $\rho < 1/2$. Thus the entire seed stays below r_0 . Its length $2H$ is fixed independently of n .

With $r_j = q^j r_0$, successively require the lower crossing events (7.3). Each ends in J and takes at most Tr_{j+1}^2 steps. The mass of successful prefixes through radius r_J is at least the positive seed probability times

$$q^{-(p+\eta)J}, \quad (8.1)$$

and their total time is at most

$$2H + T \sum_{j=1}^J r_j^2 \leq 2H + \frac{Tr_J^2}{1 - q^{-2}}. \quad (8.2)$$

Choose a fixed $\delta > 0$ so small that $T\delta^2/(1 - q^{-2}) \leq 1/8$. For each sufficiently large integer n , choose the largest J with $r_J \leq \delta\sqrt{n}$. Then $J \geq 1$ and

$$\delta\sqrt{n}/q < r_J \leq \delta\sqrt{n}.$$

For all sufficiently large n , (8.2) is at most $n/4$.

These accepted prefixes stop at their first hit of the final radius r_J . This follows from the first-hit crossing construction, the no-skip condition, and the seed bound. Their scaled terminal points belong to one fixed compact subset A_0 of the open quadrant: after whitening, their radii are between δ/q and $\delta + o(1)$ and their angles lie in J . Enlarge this compact set slightly to cover all sufficiently large n .

Let F_n and G_n be the total masses of accepted original and dual prefixes, respectively. By (8.1),

$$F_n \geq c_\eta n^{-(p+\eta)/2}, \quad G_n \geq c_\eta n^{-(p+\eta)/2}. \quad (8.3)$$

Their actual integer stopping times, at most $n/4$, are retained throughout. No random-time replacement has occurred.

8.2 The upper endpoint estimate

Split n into three integer parts $n_1 + n_2 + n_3 = n$, each comparable to n . Drop killing in the middle part and apply (4.3):

$$\begin{aligned} K_n((0, c), (0, W)) &\leq \frac{C}{n} \left(\sum_{x,a} K_{n_1}((0, c), (x, a)) \right) \left(\sum_{y,b} K_{n_3}((y, b), (0, W)) \right) \\ &\leq \frac{C'}{n} s_{n_1}(c) s_{n_3}^*(W). \end{aligned}$$

The second step uses (3.11) and the bounded factor π_W/π_b . Applying (7.7) with half the desired error gives, for every $\varepsilon > 0$,

$$\max_c K_n((0, c), (0, W)) \leq C_\varepsilon n^{-p-1+\varepsilon}. \quad (8.4)$$

8.3 The lower endpoint estimate and injectivity

For each accepted original prefix of length $t \leq n/4$ ending at (x, a) and each accepted dual prefix of length $u \leq n/4$ ending at (y, b) , insert an original-chain quadrant bridge between these states of exact integer length

$$m = n - t - u \in [n/2, n].$$

Lemma 5.1 gives total bridge probability at least c/n , uniformly in these data. Reverse the dual prefix to form the final suffix ending at $(0, W)$. Its original probability is π_W/π_b times its dual probability, bounded below by a positive constant times that probability.

There is no overcounting in summing over all such triples. In a resulting full path, the first forward hit of r_J uniquely recovers its accepted original prefix. The first backward hit of r_J , read from the terminal origin, uniquely recovers its accepted dual prefix. These pieces cannot overlap because $t, u \leq n/4$. Their colors, endpoints, stopping times, and complete histories are fixed by the full path, and the middle segment is then fixed as well. Further radial crossings in the bridge do not change either recovered first hit.

Summing over actual stopping times, positions, colors, and prefix histories consequently gives

$$K_n((0, B), (0, W)) \geq \frac{c}{n} F_n G_n \geq c_\eta n^{-p-1-\eta}. \quad (8.5)$$

Together (8.4) and (8.5) imply

$$K_n((0, B), (0, W)) = n^{-p-1+o(1)}.$$

The same logarithmic exponent holds for the positive finite weighted sum in (3.12): only the black start is needed for its lower bound, while all colors satisfy the upper bound. Finally $L = n - 1$ changes Γ^L by a constant factor and $\log L$ by $o(1)$. This proves (1.2).

Order of constants. The proof fixes an exponent tolerance η first; then a sufficiently large $q \geq 2$ absorbing the Brownian crossing constants; then a time cutoff $T(q)$ and a least radius $R_0(q)$ including the no-skip bound; then the integer seed height H and r_0 ; and finally δ . Only after all these are fixed does n tend to infinity. The functional limit is never required to be uniform as $q \rightarrow \infty$.

9 The threshold inverse

The sequence is nondecreasing. For any excursion starting in color c , prepend a zero-displacement edge $R \rightarrow c$, available for every c . This produces an excursion one step longer; deleting its first edge recovers the original, so the map is injective. Also $a(0) = a(1) = 1$. By the exponential growth theorem the sequence is unbounded, so $N(y)$ is well defined.

[write] **Added 5 October 2026.** In fact $a(n+1) \geq 2a(n)$ for $n \geq 1$, so the sequence is strictly increasing from $n = 1$ on $(1, 2, 6, 20, \dots)$. Proof: an excursion counted by $a(n)$ has length $n - 1 \geq 0$ and starts at the origin in some colour c . Prepending either the zero-displacement edge $R \rightarrow c$ or the zero-displacement edge $G \rightarrow c$ (both exist for every c : rows R and G of the table) gives an excursion of length n , counted by $a(n+1)$. Each map is injective (delete the first edge), and their images are disjoint, since the new initial colours differ. The inverse below uses only monotonicity.

Put $g = \log \Gamma$ and $L = \log y$. From (1.2), $\log a(n)/n \rightarrow g > 0$, and monotone bracketing gives $N(y) \sim L/g$. More explicitly, with $N = N(y)$ and $y > 1$, minimality gives

$$\log a(N-1) < L \leq \log a(N).$$

Apply the already proved asymptotic separately at N and at $N-1$. Since $N \rightarrow \infty$, $(N-1)g = Ng + O(1)$ and $\log(N-1) = \log N + o(1)$, both sides have the form

$$gN - \alpha \log N + o(\log N).$$

Consequently

$$L = gN - \alpha \log N + o(\log N), \quad \log N = \log L - \log g + o(1).$$

Substitution yields

$$N = \frac{L}{g} + \frac{\alpha}{g} \log L + o(\log L),$$

which is (1.3). The constant contribution $-(\alpha/g) \log g$ lies below the proved error scale. It cannot be claimed as a resolved bounded term. The argument does not differentiate the $o(\log n)$ remainder or assume a bound on its finite differences.

Remark 9.1 ([write] A Lambert form of the inverse). The model equation $gX - \alpha \log X = L$ is the equation $aX + b \log X = L$ of `p0:thm:lambert-core` in the repository's transseries volume (path in Section 1.1) with $a = g$ and $b = -\alpha < 0$. Its branch rule gives, for L above the threshold $\alpha(1 - \log(\alpha/g))$, a unique solution larger than α/g ,

$$X_0(L) = -\frac{\alpha}{g} W_{-1}\left(-\frac{g}{\alpha} e^{-L/\alpha}\right) = -\frac{\alpha}{g} W_{-1}\left(-\frac{g}{\alpha} y^{-1/\alpha}\right),$$

and the proof above shows $N(y) = X_0(\log y) + o(\log \log y)$. Indeed, it gives $\Phi(N) = L + e$ with $\Phi(X) = gX - \alpha \log X$ and $e = o(\log N)$, while $\Phi(X_0(L)) = L$; on $[2\alpha/g, \infty)$, which contains N and $X_0(L)$ for large y , $\Phi' \geq g/2$, so $|N - X_0(L)| \leq 2|e|/g = o(\log N) = o(\log L)$. Expanding X_0 gives back (1.3); at this precision the Lambert form adds no information, and no novelty is claimed for the inversion mechanics. The bounded term $-(\alpha/g) \log g$ of X_0 is still below the error scale, as said above.

10 Non D finiteness from the logarithmic power

10.1 An irrational exponent between two and three

The exact inequalities $-1 < (-29 + 7\sqrt{17})/4 < 0$ show $\pi/2 < \theta < \pi$, hence $2 < \alpha < 3$. Furthermore

$$u = 2 \cos \theta = \frac{-29 + 7\sqrt{17}}{2}$$

has conjugate $u' = (-29 - 7\sqrt{17})/2 < -2$. It satisfies the irreducible quadratic $u^2 + 29u + 2 = 0$, whose discriminant is $833 = 49 \cdot 17$, not a rational square. If θ/π were rational, $e^{i\theta}$ would be a root of unity and every algebraic conjugate of $e^{i\theta} + e^{-i\theta}$ would belong to $[-2, 2]$. The conjugate u' contradicts that property. Thus θ/π , $p = \pi/\theta$, and α are irrational. This is an exact algebraic certificate, not a decimal test.

10.2 A real singular exponent forced by positive coefficients

The radius of convergence of F is $R = \Gamma^{-1}$. Define $H(t) = F(Rt)$ for $0 < t < 1$. Equation (1.2) gives

$$a(n)R^n = n^{-\alpha+o(1)}, \quad H''(t) = \sum_{n \geq 2} n(n-1)a(n)R^n t^{n-2}.$$

For every sufficiently small $\varepsilon > 0$, the coefficients of this positive series are, for all sufficiently large n , between fixed positive multiples of $n^{2-\alpha-\varepsilon}$ and $n^{2-\alpha+\varepsilon}$. Both exponents exceed -1 if $\varepsilon < 3 - \alpha$.

For each fixed real $s > -1$,

$$\sum_{n \geq 1} n^s t^n \asymp (1-t)^{-s-1} \quad (t \uparrow 1). \quad (10.1)$$

To prove the comparison, write $t = e^{-u}$. Terms with $n \in [1/u, 2/u]$ give the lower bound. For the upper bound, split into the initial block $n \leq 1/u$ and successive blocks $n \in [k/u, (k+1)/u]$, $k \geq 1$. The initial bound uses integrability of x^s at zero; the other blocks sum a convergent exponentially decaying series times u^{-s-1} . Finally $u \sim 1 - t$. Removing finitely many terms does not affect the comparison.

Let $\beta = 3 - \alpha > 0$. Positivity and (10.1) give

$$c_\varepsilon (1-t)^{-\beta+\varepsilon} \leq H''(t) \leq C_\varepsilon (1-t)^{-\beta-\varepsilon}$$

for t near one. In particular $H''(t) \rightarrow +\infty$, and division of logarithms, followed by $\varepsilon \downarrow 0$, gives

$$\lim_{z \uparrow R} \frac{\log F''(z)}{\log(R-z)} = \alpha - 3 \notin \mathbb{Q}. \quad (10.2)$$

Here $H''(t) = R^2 F''(Rt)$; fixed factors and the linear change of variable do not alter the limit. This uses only the logarithmic coefficient power, not a coefficient equivalent.

10.3 The arithmetic obstruction

Suppose that F were D-finite over $\mathbb{Q}(z)$. Its coefficients are integers with exponential growth, by (3.12); all their algebraic conjugates obey the same bound and their common denominators equal one. Thus it satisfies the definition of a G-function. The same is true of F'' , whose coefficients are $(n+2)(n+1)a(n+2)$ and remain exponentially bounded integers.

The arithmetic local-structure theorem for G-functions says that their minimal differential operators are regular singular with rational exponents. We use the formulation in Fischler and Rivoal [4, Definition 1, pp. 313–314; Theorem 6, pp. 328–329; Corollary 1, p. 329]. At an algebraic point, on a chosen local branch, a solution is a finite sum of terms

$$w^q (\log w)^k f(w), \quad q \in \mathbb{Q}, \quad k \in \mathbb{Z}_{\geq 0},$$

where f is holomorphic at zero. This is the André–Chudnovsky–Katz arithmetic theorem in that source. Here the relevant algebraic point is R and we take $w = R - z$ along the real interval $z < R$ on the original power-series branch. Any operator singularities inside the convergence disk cause no obstruction: this particular power-series solution is analytic through them, even if other solutions are not. A sufficiently small punctured neighborhood of R contains no other operator singularity; choose its slit so that the real interval to the left of R lies in the branch domain.

Combining like rational exponents and taking the first nonzero Taylor coefficients gives a leading term

$$F''(z) = c(R - z)^q (\log(R - z))^k (1 + o(1)), \quad q \in \mathbb{Q}, \quad k \geq 0, \quad c \neq 0. \quad (10.3)$$

To see why cancellation is harmless, group exponents whose differences are integers, combine their holomorphic coefficients and logarithmic polynomials, and then take the least remaining power and its largest nonzero logarithmic degree. The finite collection has a first nonzero term. This consequence is also stated in the cited Corollary 1. Reality and positivity of F'' on this interval select a positive real leading sign after the factor $(-1)^k$ is absorbed. Equation (10.3) implies

$$\lim_{z \uparrow R} \frac{\log F''(z)}{\log(R - z)} = q \in \mathbb{Q},$$

contradicting (10.2).

If D-finiteness were initially allowed over $\mathbb{C}(z)$, clear rational-function denominators and fix the finite order and polynomial degrees of a nonzero annihilating operator. Its unknown coefficients solve homogeneous linear equations with rational entries, because the coefficients of F are rational. This infinite system has a finite-dimensional row space and therefore a finite rational basis of rows. A nonzero complex solution entails a nonzero rational solution by rational row reduction. Thus D-finiteness descends to $\mathbb{Q}(z)$ and is already excluded.

Bostan, Raschel and Salvy [5, Theorem 3] use the same arithmetic mechanism with a full coefficient equivalent. That criterion is not directly applied here: the positive-second-derivative comparison above supplies the extra argument needed for the weaker information $n^{-\alpha+o(1)}$. Nor is any claim being made about arbitrary nonarithmetic D-finite functions.

Remark 10.1 ([write] The criterion in general). The argument of this section proves: *if a_n are nonnegative integers, $\mu > 1$, α is irrational and $a_n = \mu^n n^{-\alpha+o(1)}$, then $\sum_n a_n z^n$ is not D-finite over $\mathbb{C}(z)$.* Take for k the least nonnegative integer with $k > \alpha - 1$, and replace F'' and H'' by $F^{(k)}$ and $H^{(k)}$. The coefficients of $H^{(k)}$ are nonnegative and, for large n , between fixed multiples of $n^{k-\alpha \mp \varepsilon}$, with $k - \alpha - \varepsilon > -1$ for small ε ; (10.1) gives $H^{(k)}(t) \asymp_\varepsilon (1 - t)^{-(k+1-\alpha) \pm \varepsilon}$ with $k + 1 - \alpha > 0$, so $H^{(k)}(t) \rightarrow \infty$ (finitely many omitted terms add a bounded polynomial) and $\log F^{(k)}(z) / \log(R - z) \rightarrow \alpha - k - 1 \notin \mathbb{Q}$, $R = \mu^{-1}$. The coefficients of $F^{(k)}$, $(n+k) \cdots (n+1)a_{n+k}$, are integers with exponential bounds, so $F^{(k)}$ is a G-function if F is D-finite, and the rest of the argument, including the descent from $\mathbb{C}(z)$ to $\mathbb{Q}(z)$, is unchanged. Since $\Theta(\mu^n n^{-\alpha})$ implies $\mu^n n^{-\alpha+o(1)}$, Lemma `cps:lem:nonD` of `a377922-corner-polyhedra-schnyder` (Part I) is the special case with two-sided bounds. Report 112 (Part III there, its Section 5) runs the same argument with $k = \lfloor \alpha \rfloor$ for its two sequences.

11 Validation and remaining questions

11.1 What can be checked finitely

The accompanying source-checks package supplies a deterministic build and exact finite checks, with explicit runtime guards that remain enabled under Python optimization. It verifies the walk prefix against the 17 source terms, independently enumerates vincular pattern avoiders, checks Perron and stationary equations and the corrector covariance in $\mathbb{Q}(\sqrt{17})$, and verifies reversal, aperiodicity paths, seed and connector steps, annular scheduling inequalities, and the algebraic irrationality certificate. Deliberately corrupted inputs are required to fail. A strict manifest rejects missing,

changed, or unexpected payload files; a fresh extraction is replayed through verification and PDF compilation.

These checks establish reproducibility of the finite data and symbolic identities. They cannot certify the functional central limit theorem, harmonic measure, uniform annular transfer, bridge tightness, or the asymptotic conclusion by numerical testing. Those steps are proved in the text. The source-check record identifies the supporting primary references and exact locations; third-party source PDFs are not redistributed.

[write] **Shipped layout and intake reruns, added 5 October 2026.** In ProveIt the package is shipped renamed (the report README lists every file): the checker, the mutation campaign and the packaging scripts are in `code/`, the fixture and recorded results in `data/`, the checks README as `checks-README.md`. The integrity, replay and sealing scripts need the delivered `manifest.json` and layout, and are run from the archive of the arrival commit. On copies (Windows, Python 3.14.4), the intake found: the delivered checksums and manifest verified; the checker, normal and optimized, reproducing the recorded result after removal of the carriage returns that Windows redirection adds (394,716 guards); the mutation campaign (54 mutations, 108 rejected runs) and the integrity corruption test (24 rejections) passing, with outputs identical to the recorded ones up to those carriage returns. The delivered `build.py` assumes a TeX Live format build and was not run; this PDF is built with MiKTeX pdfL^AT_EX. An independent intake script reconstructed the step table from the two weak inequalities, recomputed $a(0), \dots, a(16)$, A , its characteristic polynomial, r , π , m , h , Σ (also as the Hessian of $\log \lambda$, by finite differences), ρ , α and the quadratic of u ; all agree.

11.2 Bounded literature scope

The relevant published comparison is the 2025 source [2]; its Proposition 4.23 and following conjecture are distinguished explicitly above. A targeted literature review through 2 October 2026 did not locate a later proof resolving this same logarithmic statement. This is a bounded search, not an assertion that all literature or unpublished work has been exhausted. The result should be compared with subsequent specialist work before any publication or priority claim.

11.3 Concrete next mathematical questions

1. Prove the full coefficient equivalent (1.4). This would require finer control than fixed-error annular powers, for example a valid finite-state killed-cone local limit theorem with the appropriate endpoint harmonic functions.
2. Identify positive forward and dual discrete harmonic functions, normalize them consistently, and express or compute the proposed amplitude c . The present proof discards fixed crossing constants and cannot determine it.
3. Obtain a quantitative error in the logarithmic asymptotic. A rate in the annular transfer or an exact killed-kernel theorem could potentially replace $o(\log n)$ by a more informative remainder.
4. Determine whether correction powers, logarithms, or additional singular contributions occur. The covariance angle predicts the leading logarithmic power only; no correction hierarchy follows from it.
5. If a full equivalent with amplitude becomes available, refine the threshold inverse to its bounded term. Any claimed smaller inverse error must also handle the integer rounding inherent in $N(y)$.

[write] **Added 5 October 2026.** These five questions are restated, with what is missing, as Questions 1–5

of Section 11.4.

11.4 Further questions and research

[write] **Added 5 October 2026, batch 103.** Following Vladimir Reshetnikov's standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. Questions 1–5 are the five items above; Question 6 collects the two condensed steps that the write did not complete; Question 7 concerns the external inputs. The intake found no false claim in the source.

1. **The full equivalent** (source, item 1; the amplitude part of the conjecture (1.4) of [2]). Prove $a(n) \sim c\Gamma^n n^{-\alpha}$ with $c > 0$. The source suggests a finite-state killed-cone local limit theorem with endpoint harmonic functions. Missing: every estimate finer than $n^{o(1)}$; the annular argument of Section 7 loses a factor $q^{\pm\eta}$ per scale and cannot give one. A model for this step: Report 125 of the same bundle (A279571, inversion sequences avoiding 100, 101, 201; archive A279571_Leading_Equivalent_and_Harmonic_Constant_Source.zip, arrival 60f54ea06, to be filed by a later intake batch) proves a leading equivalent $a_n \sim C_A 9^n n^{-\kappa}$, $\kappa = 1 + \pi / \arccos(2/\sqrt{7})$, for an exact two-colour quadrant walk of the same kind, with a killed local limit theorem built from normalized forward and reversed killed harmonic functions, a deep-entrance argument, a side-uniform Brownian limit and midpoint reversal. Whether that route carries over to the four colours and the origin endpoint here is not checked.
Observation, not a claim. An intake floating-point computation of the recurrence (2.2) (double precision with renormalization, not interval arithmetic) gives $a(n)\Gamma^{-n}n^\alpha = 4.154, 4.418, 4.557, 4.604, 4.627, 4.641, 4.651$ at $n = 50, 100, 200, 300, 400, 500, 599$, still increasing slowly, and secant exponents $-\log(a(n)\Gamma^{-n}/a(m)\Gamma^{-m})/\log(n/m)$, $m = \lfloor n/2 \rfloor$, of 2.782, 2.868, 2.912, 2.927, 2.935, 2.939, 2.942, approaching $\alpha = 2.9569$ from below. This is consistent with the conjecture and certifies nothing; no value of c is claimed.
2. **Harmonic functions and the amplitude** (source, item 2). Identify positive forward and dual discrete harmonic functions for the killed chain, normalize them consistently, and express or compute c . Missing: their existence and asymptotics (in Report 125 these are the limits V, V^* with $V = U + O((1 + |s|)^{p-1+\delta})$, U the wedge harmonic function there); the present proof discards every fixed crossing constant.
3. **A quantitative error term** (source, item 3). Replace $o(\log n)$ in (1.2) by an explicit rate. Missing: a rate in the annular transfer (7.2), (7.3), which rests on the qualitative functional limit, or an exact killed-kernel theorem.
4. **Corrections and further singular contributions** (source, item 4). Do correction powers, logarithms or other singular contributions occur? The covariance angle predicts the leading power only; nothing in the text bears on the correction hierarchy.
5. **A bounded-term inverse** (source, item 5). With an amplitude, refine (1.3) to its bounded term, the term $-(\alpha/g)\log g$ of Remark 9.1 plus an amplitude shift, and handle the integer rounding in $N(y)$. Missing: Question 1, and a rounding analysis.
6. **The condensed steps 3 and 4** (Lemma 5.1; uniformity in Section 7). Write out the bridge limit and its tightness through the dual chain, and the contradiction and portmanteau arguments of the annular transfer, as the notes after Lemma 5.1 and at the end of the transfer subsection describe. The intake judged both correct in outline; Report 112's Theorem 2.1 (Remark 1.2) proves the same exponent for a class containing this chain, with its own versions of these steps (its Lemma on exact-time interior bridges and its crossing bounds), which the

intake did not check line by line either. Steps 1 and 2 (after Lemma 3.1 and in Section 4) are completed by the write.

7. **External inputs and priority.** The bijection of [2] and its weak, rather than strong, leftmost and rightmost conditions; the cited statements of [3] and [4] and their page locations; and the source's bounded literature search (through 2 October 2026; an intake web search on 5 October 2026 found no later proof either). None was checked against the primary sources by ProveIt. The source asks that the result be compared with subsequent specialist work before any publication or priority claim.

References

- [1] The OEIS Foundation Inc. *A348351*, number of permutations avoiding four specified vincular patterns. Offset 0; terms through $n = 16$ checked 2 October 2026. <https://oeis.org/A348351>.
- [2] A. Asinowski, J. Cardinal, S. Felsner and É. Fusy. *Combinatorics of rectangulations: Old and new bijections*. Combinatorial Theory **5**(1) (2025), article 14. DOI: 10.5070/C65165025. Section 4.6, printed pp. 39–40, Figure 4.16, Proposition 4.23 and following paragraph. <https://escholarship.org/uc/item/92s0c9qg>.
- [3] W. Whitt. *Proofs of the martingale FCLT*. Probability Surveys **4** (2007), 268–302. DOI: 10.1214/07-PS122. Theorem 2.1(ii), pp. 270–271. <https://www.columbia.edu/~ww2040/WWproofs2007pub.pdf>.
- [4] S. Fischler and T. Rivoal. *On the values of G -functions*. Commentarii Mathematici Helvetici **89** (2014), 313–341. DOI: 10.4171/CMH/321. Definition 1, pp. 313–314; Theorem 6, pp. 328–329; Corollary 1, p. 329. <https://ems.press/content/serial-article-files/43350>.
- [5] A. Bostan, K. Raschel and B. Salvy. *Non- D -finite excursions in the quarter plane*. Journal of Combinatorial Theory, Series A **121** (2014), 45–63. Theorem 3 is discussed for comparison, not invoked as a full-equivalent criterion here. <https://specfun.inria.fr/bostan/publications/BoRaSa14.pdf>.